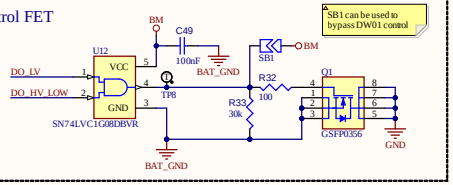
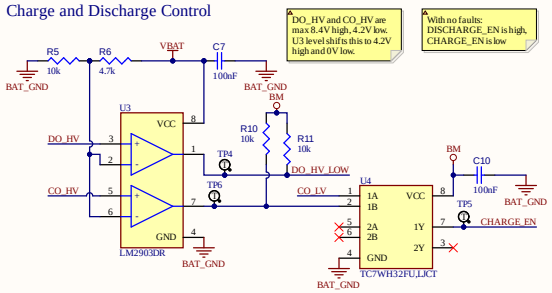
Battery Management

In an attempt to reduce costs I present the world's worst BMS ever

Discharge Control FET



Charge and Discharge Control

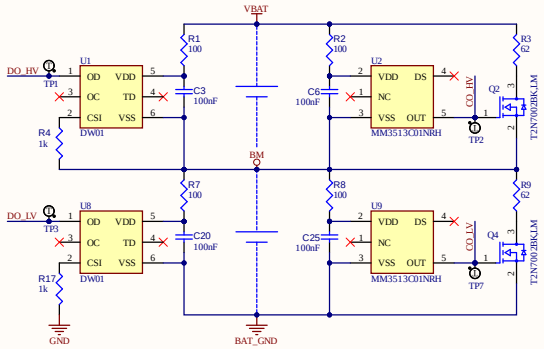


Cell Monitoring and Balancing

These trip at 2.4V, so they are only a safeguard. The Raspberry Pi should shut off the robot around 3V. At 2.4V the cells would have a high likelihood of damage.

Overcurrent protection is a minimum of 13A or so, realistically more like 18A.

The MM3513 is used for balancing since the DW01's overvoltage threshold and hysteresis is too high.

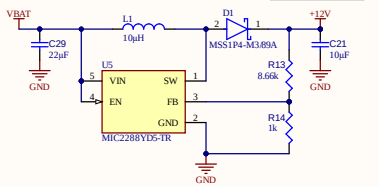


For overcurrent monitoring, the voltage drop over Q1 (GND to BAT_GND) is measured by U8.

DC-DC Converters

12V Boost Converter

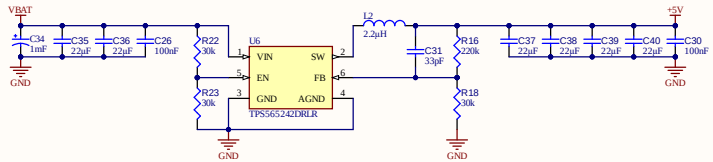
Good for about 600mA



5V Buck Converter

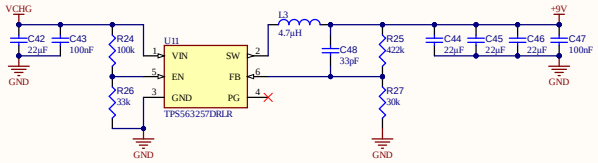
This thing has a comical amount of capacitors

Up to 5A output



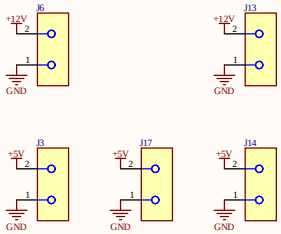
9V Buck Converter

Up to 2.5A output

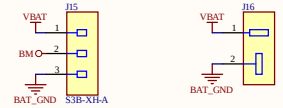


Connectors

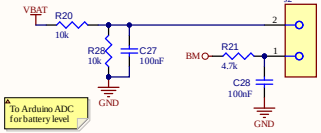
Power Outputs



Battery Input

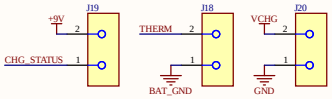


Battery Monitor Output

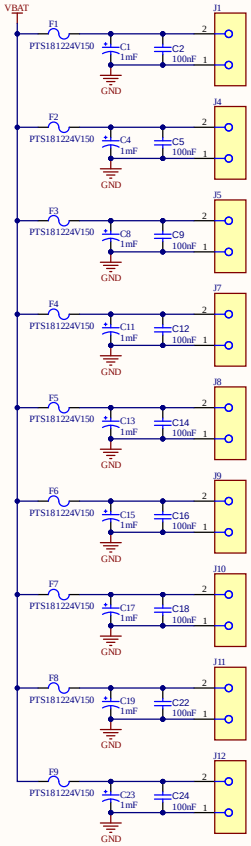


To Arduino ADC for battery level

Charger I/O

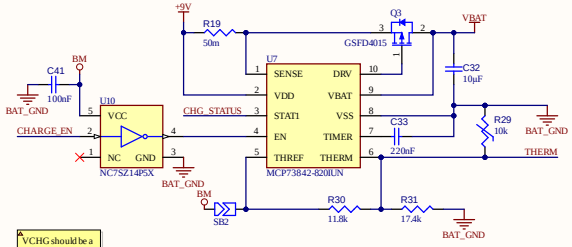


Servo Outputs



Charger

Max input current: ~2.2A, leading to ~8.8W of power dissipation in Q1



VCHG should be a 12V external DC power supply, able to supply 2.5A or more of continuous current

SB2 can be used if we want to monitor a thermistor off-board

Replace R30 with a 10k and R31 with a 6.8k to disable temperature monitoring

Title: Ripple Robot Power Management System		
Size: C	Number	Revision: A
Date: 4/16/2026		Sheet 1 of 1
File: C:\Users\..\Ripple_PMS.SchDoc		Drawn By: Graham Shupe

+12V
GND
+5V
GND
+5V
GND
MONITOR
M
9
SERVO
8
SERVO
7
SERVO
6
SERVO
5
B+
B−
1
SERVO
2
SERVO
3
SERVO
4
CHARGER
BATTERY
THERM
STATUS